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Chair of Aerospace Information Technology

Online Floatsat 2021
Attitude Determination – Part 3

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What are we going to learn?

Part3: Complex Filter Design and Implementation

- Understanding and implementing of two more complex filters

  → Mahony Filter and Madgwick Filter

Attitude Determination – Gyroscope

Gyroscope measures the angular velocity in 3 axis

$$\omega = \left[\omega_x \; \omega_y \; \omega_z\right]^T = \left[\dot{\phi} \;\; \dot{\theta} \;\; \dot{\psi}\right]^T \qquad \Omega = \left[0 \;\; \omega_x \; \omega_y \; \omega_z\right]^T$$

→ Update the quaternion with the gyroscope

$$q_{\mathrm{w,t}} = q_{t-1} + \frac{\Delta t}{2}\left(\Omega_t \otimes q_{t-1}\right) = \begin{bmatrix} 1 & -\frac{\Delta t}{2}\omega_x & -\frac{\Delta t}{2}\omega_y & -\frac{\Delta t}{2}\omega_z \\ \frac{\Delta t}{2}\omega_x & 1 & \frac{\Delta t}{2}\omega_z & -\frac{\Delta t}{2}\omega_y \\ \frac{\Delta t}{2}\omega_y & -\frac{\Delta t}{2}\omega_z & 1 & \frac{\Delta t}{2}\omega_x \\ \frac{\Delta t}{2}\omega_z & \frac{\Delta t}{2}\omega_y & -\frac{\Delta t}{2}\omega_x & 1 \end{bmatrix} \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} = \begin{bmatrix} q_w - \frac{\Delta t}{2}\omega_x q_x - \frac{\Delta t}{2}\omega_y q_y - \frac{\Delta t}{2}\omega_z q_z \\ q_x + \frac{\Delta t}{2}\omega_x q_w - \frac{\Delta t}{2}\omega_y q_z + \frac{\Delta t}{2}\omega_z q_y \\ q_y + \frac{\Delta t}{2}\omega_x q_z + \frac{\Delta t}{2}\omega_y q_w - \frac{\Delta t}{2}\omega_z q_x \\ q_z - \frac{\Delta t}{2}\omega_x q_y + \frac{\Delta t}{2}\omega_y q_x + \frac{\Delta t}{2}\omega_z q_w \end{bmatrix}$$

Data combination with Quaternion

- **Transfer the pitch and roll angle measurements from the accelerometer and the yaw from the magnetometer to a quaternion:**

$$\mathbf{q}_{am} = \begin{pmatrix} q_w \\ q_x \\ q_y \\ q_z \end{pmatrix} = \begin{pmatrix} \cos\left(\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) + \sin\left(\frac{\phi}{2}\right)\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right) \\ \sin\left(\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) - \cos\left(\frac{\phi}{2}\right)\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right) \\ \cos\left(\frac{\phi}{2}\right)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) + \sin\left(\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right) \\ \cos\left(\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right) - \sin\left(\frac{\phi}{2}\right)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) \end{pmatrix}$$

Data combination with Quaternion

- **Final combination for the complimentary filter:**

$$\mathbf{q_t} = (1 - \beta) \cdot q_{w,t} + \beta \cdot q_{am,t}$$

- **To use the quaternion for the attitude we need to normalize it!**

$$\hat{\boldsymbol{q}} = \frac{\mathbf{q}}{||\boldsymbol{q}||}$$

Complimentary Filter

- Easy implementation and fast runtime
- But: We are ignoring the process noise, bias and other errors

  → Attitude error is ignored and will drift over time
- Idea: Compare the measured attitude with the estimated attitude
- Solution: Usage of more complex filters

  → i.E. Mahony Filter, Madgwick Filter

Madgwick Filter: Introduction

- Based on a publication of Madgwick et al.

  → https://ieeexplore.ieee.org/document/5975346

- Compare the measured attitude with the estimated attitude

- Idea: Formulate an optimization problem to minimize the error between predicted and measured orientation

  → Use a gradient descent algorithm to solve this problem

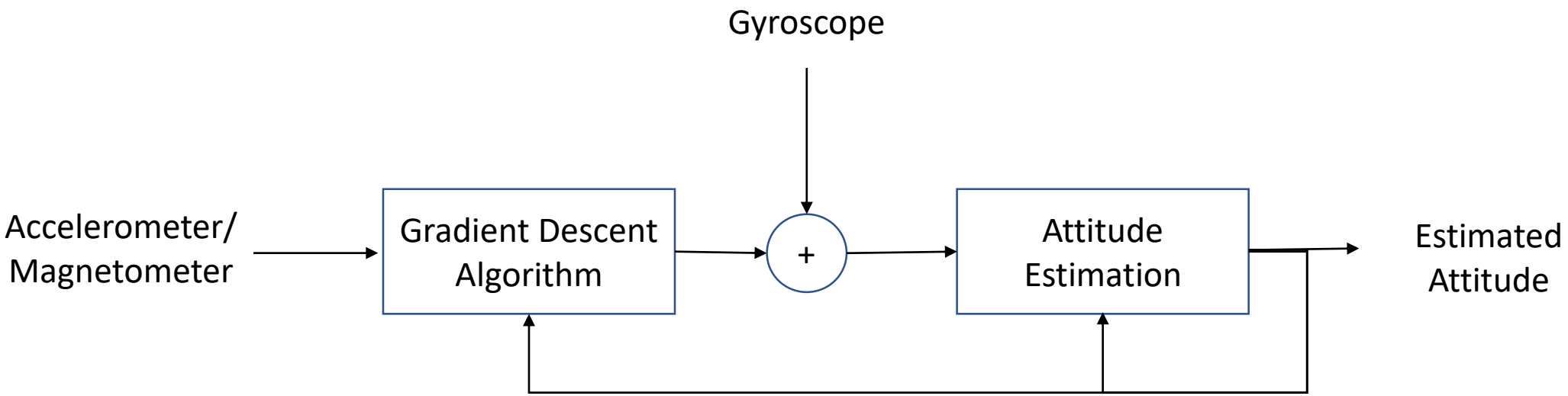
Madgwick Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation

1. **Optimization Problem: Minimize error between estimated and measured attitude (accelerometer/magnetometer)**
2. **Use a Gradient Descent Algorithm to compute the solution to this problem**
3. **Propagate/Estimate the attitude using the corrected gyroscope**
4. **Normalize attitude quaternion and go back to step 1**

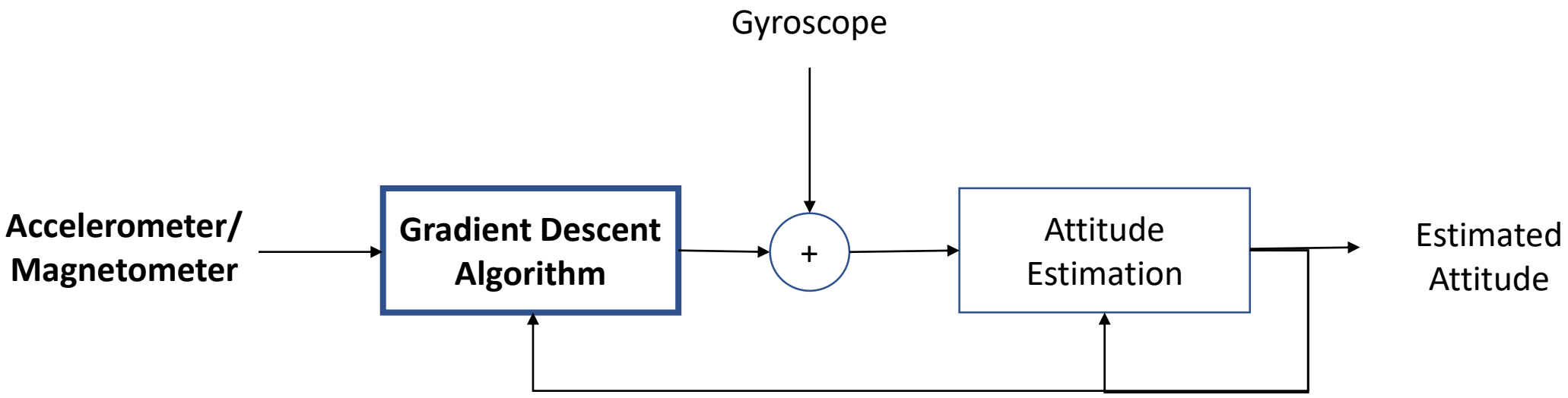
Madgwick Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation



Madgwick Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation



Madgwick Filter: Gradient Descent Algorithm

We define:

- **A predefined reference** $d = \left[0\ d_x\ d_y\ d_z\right]^T$
- **The corresponding measurement** $\mathrm{s} = \left[0\ s_x\ s_y\ s_z\right]^T$

If our attitude estimate is perfect / without error, rotating d by our attitude estimated q will result in the corresponding measurement s

$$f(q, d, s) = q^* \cdot d \cdot q\ - s$$

Madgwick Filter: Gradient Descent Algorithm

- **If our attitude estimate is perfect / without error, rotating d by our attitude estimated q will result in the corresponding measurement s**

$$f(q, d, s) = q^* \cdot d \cdot q \; - s$$

- **Therefore our attitude q can be found as the solution to:**

$$\min f(q, d, s)$$

→ For computing the solution we can use a Gradient Descent Algorithm

Madgwick Filter: Gradient Descent Algorithm

Computing the solution by using the Gradient Descent Algorithm:

$$q_{k+1} = q_k - \mu \frac{\nabla f(q_k, d, s)}{\|\nabla f(q_k, d, s)\|}$$

with $k = 0,1,2,..,n$ **and step size** μ**, the gradient of the solution is defined as**

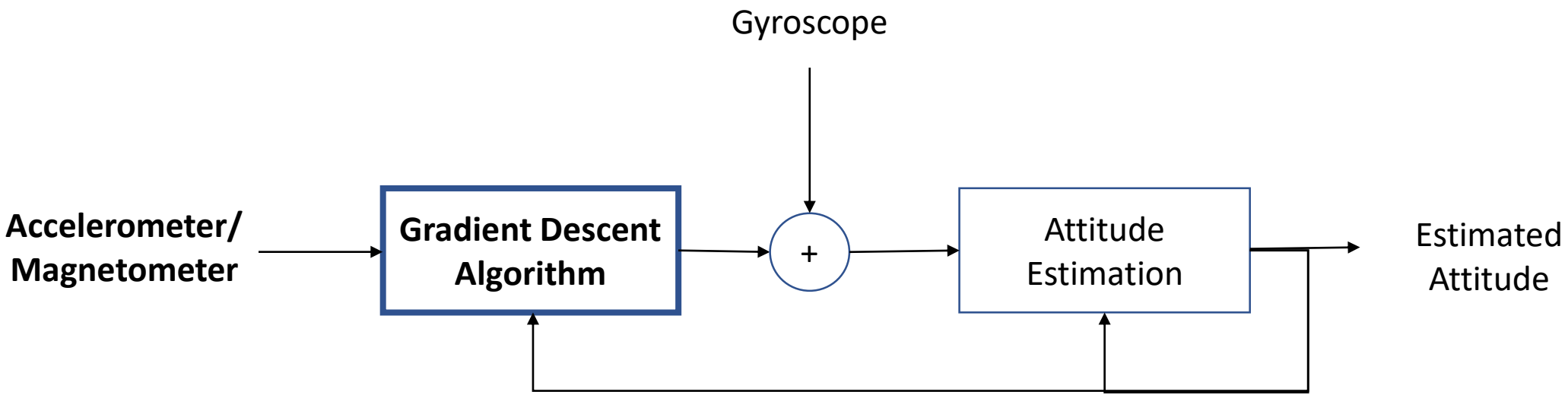
$$\nabla f(q_k, d, s) = J(q_k, d)^T \cdot f(q_k, d, s)$$

with the Jacobian of the function f:

$$J(q_k, d) = \left[\frac{\partial f(q_k, d, s)}{\partial q_w} \; \frac{\partial f(q_k, d, s)}{\partial q_x} \; \frac{\partial f(q_k, d, s)}{\partial q_y} \; \frac{\partial f(q_k, d, s)}{\partial q_z} \right]$$

Madgwick Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation



Madgwick Filter: Accelerometer

Applying the general approach to the accelerometer with:

- **Predifend Reference = gravitational force:** $g = [0\ 0\ 0\ 1]^T$
- **The corresponding measurement** $a = \left[0\ a_x\ a_y\ a_z\right]^T$

$$f_g(q,a) = q^* \cdot g \cdot q\ - \mathrm{a} = \begin{bmatrix} 2\left(q_x q_z - q_w q_y\right)\ - a_x \\ 2\left(q_w q_x + q_y q_z\right)\ - a_y \\ 2\left({}^1\!/_2 - q_x^2 - q_y^2\right)\ - a_z \end{bmatrix} \qquad J_g(q) = \begin{bmatrix} -2q_y & 2q_z & -2q_w & 2q_x \\ 2q_x & 2q_w & 2q_z & 2q_y \\ 0 & -4q_x & -4q_y & 0 \end{bmatrix}$$

Madgwick Filter: Magnetometer

Applying the general approach to the magnetometer with: (A little more complex)

- **The measured direction of the earths magnetic field in earth frame at time t**

$$h = q \cdot m \cdot q^*$$

- **The reference direction of the earths magnetic field b needs to be of the same inclination as h to reduce the effect of errors in the inclination → use only the X and Z axes of the world frame**

$$b = \left[0 \ \sqrt{h_x^2 + h_y^2} \ 0 \ h_z\right]$$

Madgwick Filter: Magnetometer

Applying the general approach to the magnetometer with: (A little more complex)

- **Predefined Reference** $b = \left[0 \ \sqrt{h_x^2 + h_y^2} \ 0 \ h_z\right]^T$
- **The corresponding measurement** $\mathrm{m} = \left[0 \ m_x \ m_y \ m_z\right]^T$

$$f_b(q, b, m) = q^* \cdot b \cdot q \ - m = \begin{bmatrix} 2b_x\left({}^1/_2 - q_y^2 - q_z^2\right) + 2b_z\left(q_x q_z - q_w q_y\right) \ - m_x \\ 2b_x\left(q_x q_y - q_w q_z\right) + 2b_z\left(q_w q_x + q_y q_z\right) \ - m_y \\ 2b_x\left(q_w q_y + q_x q_z\right) + 2b_z\left({}^1/_2 - q_x^2 - q_y^2\right) \ - m_z \end{bmatrix}$$

$$J_b(q, g, b) = \begin{bmatrix} -2b_z q_y & 2b_z q_z & -4b_x q_y - 2b_z q_w & -4b_x q_z + 2b_z q_x \\ -2b_x q_z + 2b_z q_x & 2b_x q_y + 2b_z q_w & 2b_x q_x + 2b_z q_z & -2b_x q_w + 2b_z q_y \\ 2b_x q_y & 2b_x q_z - 4b_z q_x & 2b_x q_w - 4b_z q_y & 2b_x q_x \end{bmatrix}$$

Madgwick Filter: Combine Acc/Mag

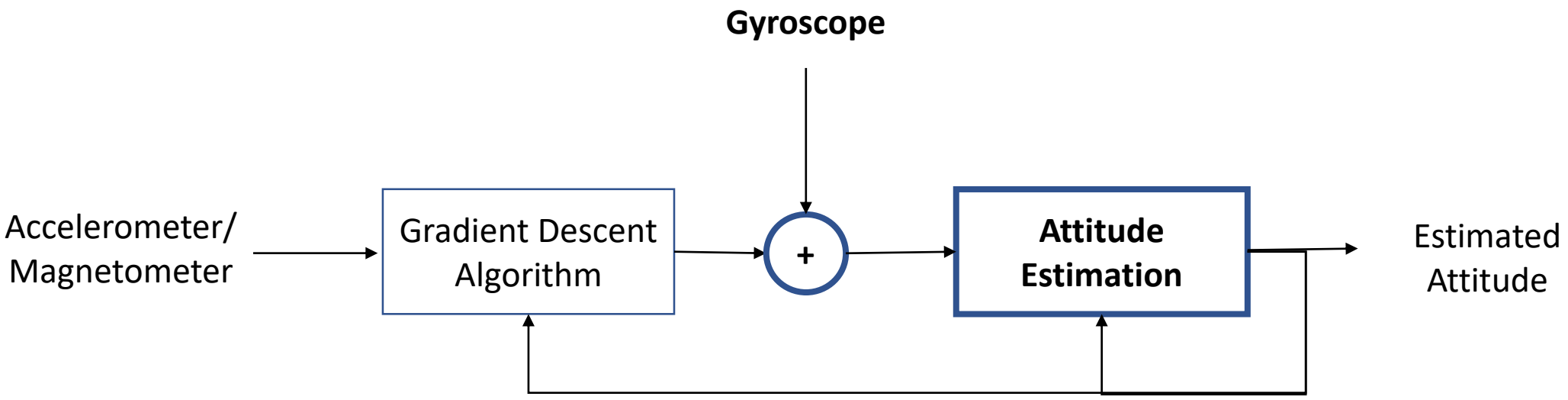
Finally combine the measurements and reference directions for both fields

$$f_{g,b}(q,a,b,m) = \begin{bmatrix} f_g(q,a) \\ f_b(q,b,m) \end{bmatrix}$$

$$J_{g,b}^T(q,b) = \begin{bmatrix} J_g^T(q) \\ J_b^T(q,b) \end{bmatrix}$$

Madgwick Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation



Madgwick Filter: Gyroscope

Gyroscope measures the angular velocity in 3 axis

$$\omega = \left[\omega_x \; \omega_y \; \omega_z\right]^T = \left[\dot{\phi} \;\; \dot{\theta} \;\; \dot{\psi}\right]^T \qquad \Omega = \left[0 \;\; \omega_x \; \omega_y \; \omega_z\right]^T$$

→ Orientation increment using the Gyroscope (same as before)

$$\dot{q}_{\mathrm{w,t}} = \frac{\Delta t}{2}\left(\Omega_t \otimes q_{t-1}\right)$$

Madgwick Filter: Attitude Estimation

Attitude Estimation similar to the complementary filter

$$q_t = q_{t-1} + \left(\dot{q}_{\omega,t} - \beta \frac{J_{g,b}^T(q_{t-1}, b)\, f_{g,b}(q_{t-1}, a, b, m)}{\left\| J_{g,b}^T(q_{t-1}, b)\, f_{g,b}(q_{t-1}, a, b, m) \right\|} \right) \Delta t$$

With the gain β as a form to represent the mean gyroscope error

→ Tunable parameter to adjust weight of gyro vs acc/mag

$$\beta = \sqrt{{}^{3}/_{4}\ \overline{\omega}}$$

Madgwick Filter: Normalize Quaternion

4. Normalize attitude quaternion and go back to step 1

Finally normalize the quaternion:

$$\hat{q} = \frac{q}{||q||}$$

Mahony Filter: Introduction

- Based on a publication of Mahony et al.

  → https://ieeexplore.ieee.org/document/4608934

- Nonlinear Complementary Filter with significant improvements in accuracy
- Idea: Compute the offset / error between predicted and measured attitude

  → Use this information to improve the next prediction

Mahony Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation

1. Compare estimated attitude with measured attitude using other sensors

   → In our case: Accelerometer and Magnetometer

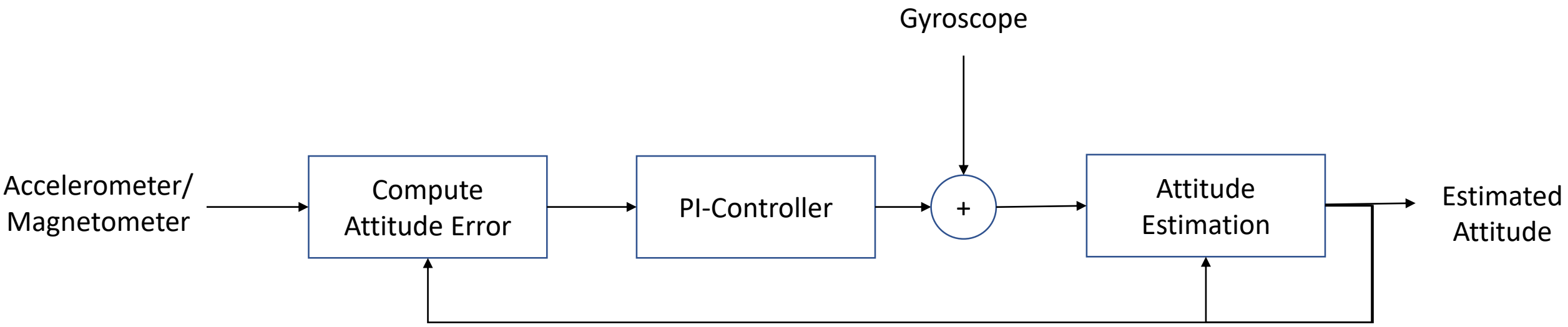
2. Use a PI-Controller to regulate/minimize the “error offset” of the gyroscope

3. Propagate/Estimate the attitude using the corrected gyroscope

4. Normalize attitude quaternion and go back to step 1

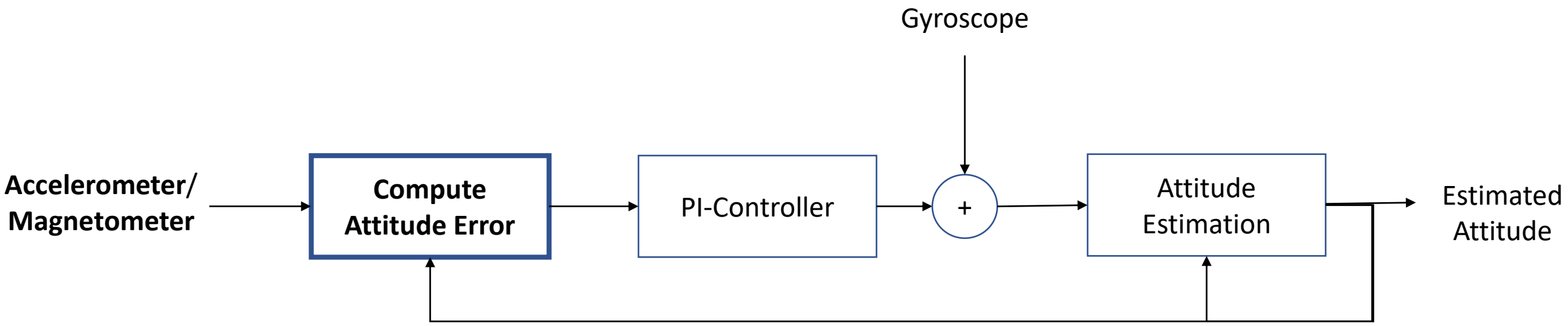
Mahony Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation



Mahony Filter: Attitude Error

Goal: Minimize the error of the attitude estimation



Mahony Filter: Attitude Error

1. Compare estimated attitude with measured attitude using other sensors

→ In our case: Accelerometer and Magnetometer

Accelerometer is measuring the gravitational force

→ Calculate the current estimated direction of the gravitational force

→ Compare this with the measured gravitational force

Mahony Filter: Accelerometer

Compare estimated direction of the gravitational field using the accelerometer

$$e_a = a \times a_{est}$$

- **If $e_a = 0$ we estimated the attitude perfectly and do not have an error**
- **The estimated acceleration a_{est} can be calculated by rotating the earths gravitational force with our current attitude quaternion q**

$$a_{est} = q^* \cdot g \cdot q = \begin{bmatrix} 2(q_x q_z - q_w q_y) \\ 2(q_w q_x + q_y q_z) \\ 2(1/2 - q_x^2 - q_y^2) \end{bmatrix}$$

Mahony Filter: Attitude Error

1. Compare estimated attitude with measured attitude using other sensors

→ In our case: Accelerometer and Magnetometer

Magnetometer is measuring the earths magnetic field

→ Calculate the current estimated direction of the earths magnetic field

→ Compare this with the measured earths magnetic field

Mahony Filter: Magnetometer

$$b = \left[0 \ \sqrt{h_x^2 + h_y^2} \ 0 \ h_z\right]$$

Compare estimated direction of the earth magnetic field using the magnetometer

$$e_m = m \times m_{est}$$

- **To calulate the estimated direction of the magnetic field m_{est} we need to rotate the magnetic field by using the attitude quaternion**

$$m_{est} = q^* \cdot b \cdot q = \begin{bmatrix} 2b_x\left({}^1/_2 - q_y^2 - q_z^2\right) + 2b_z\left(q_x q_z - q_w q_y\right) \\ 2b_x\left(q_x q_y - q_w q_z\right) + 2b_z\left(q_w q_x + q_y q_z\right) \\ 2b_x\left(q_w q_y + q_x q_z\right) + 2b_z\left({}^1/_2 - q_x^2 - q_y^2\right) \end{bmatrix}$$

Mahony Filter: Attitude Error

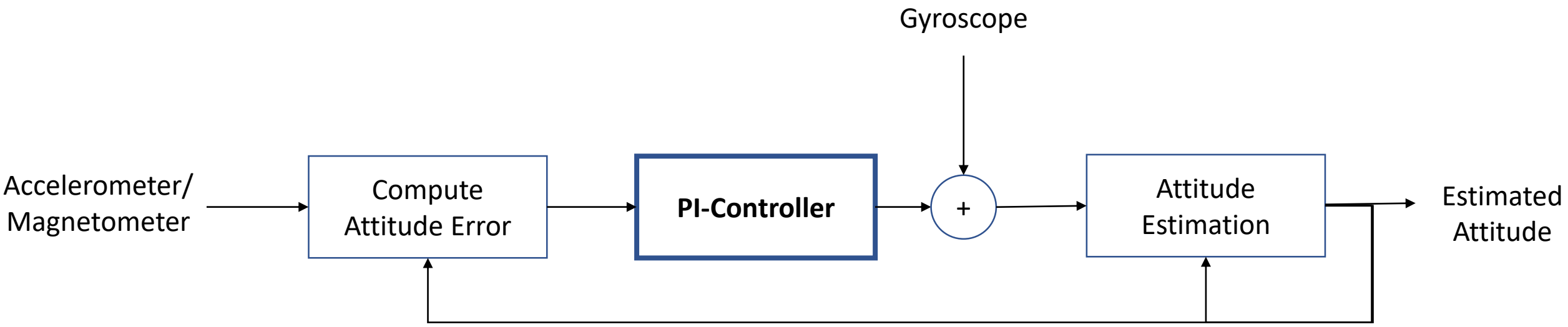
Finally: Combine both errors e_a and e_m for the attitude error

At time t the error is:

$$e_t = e_{a,t} + e_{m,t} = a_t \times a_{est,t} + m_t \times m_{est,t}$$

Mahony Filter: PI Controller

Goal: Minimize the error of the attitude estimation



Mahony Filter: PI Controller

2. Use a PI-Controller to regulate/minimize the “error offset” of the gyroscope

We calculated the error/offset using the acc/mag

→ **Use a PI-Controller with this error**

→ **Calculate a correction factor for the gyroscope**

Mahony Filter: PI Controller

- **Basic Proportional-Integral Controller**

$$u_t = k_p \cdot e_t + k_i \cdot \int_0^t e \, dt$$

- **Parameters k_p and k_i need to be determined experimently**

Mahony Filter: PI Controller

This can also be written as

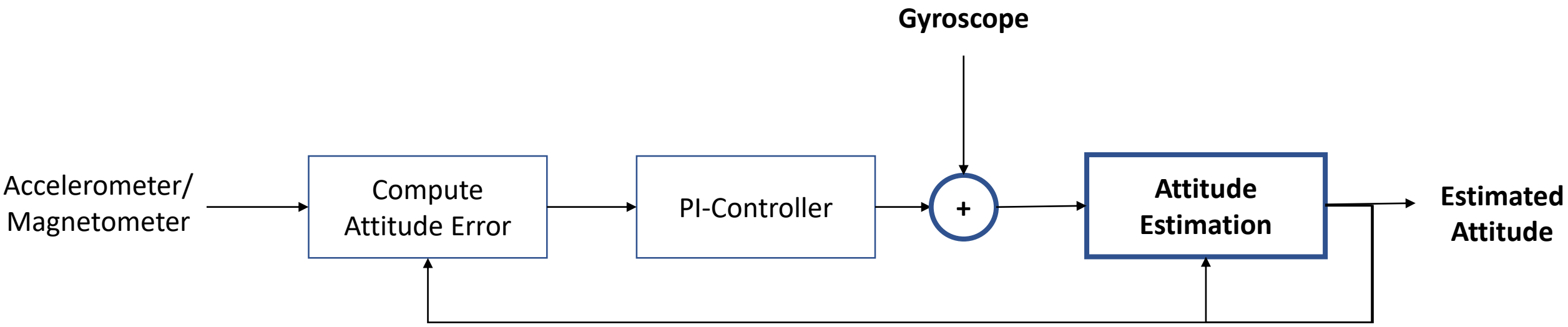
$$u_t = k_p \cdot e_t + I_t$$

with:

$$I_t = I_{t-1} + k_i \cdot e_t \cdot \Delta t$$

Mahony Filter: Principle (simplified)

Goal: Minimize the error of the attitude estimation



Mahony Filter: PI Controller

3. Propagate/Estimate the attitude using the corrected gyroscope

We calculated a correction factor for the gyroscope

→ Correct the gyroscope using the correction factor

→ Estimate the attitude using the corrected gyroscope values

Mahony Filter: Attitude Estimation

Attitude Estimation similar to the complementary filter

$$q_t = q_{t-1} + \dot{q}_{t-1}\Delta t$$

→ Using the correction factor for the gyroscope

$$q_t = q_{t-1} + \frac{\Delta t}{2} q_{t-1} \left(\Omega_t + u_t\right)$$

Mahony Filter: Normalize Quaternion

4. Normalize attitude quaternion and go back to step 1

$$\hat{q} = \frac{q}{||q||}$$

Task3: Complex Filter

Implement a more complex AHRS

- Use the formulas given in this document
- Implement the filter of your choice (Mahony or Madgwick)

  → Or implement both (its worth the effort)
- Compare all implemented filters (complementary vs quaternion vs complex filter) in realtime

Part 3: Questions

Please answer the following questions:

- **Whats the basic idea of Mahony and Madgwick to improve the solution in comparison to the simple complementary filters?**
- **What factor(s) can be adjusted in the Madgwick Filter?**
- **What factor(s) can be adjusted in the Mahony Filter?**

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